

Liquid fuels demand and supply model

Week 1 analytical pack

South Africa | Petrol and diesel

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com



Imports are material; Vopak growth depends on accessible customer flows

Current answer | validate customer access before committing additional capacity

Agenda	SCR	Current answer and implication	Next step / proposed lead	Deep dives
Market baseline	S	Imports underpin the national market 2024 net petrol/diesel imports: 13.078 bn L. Production, stocks and sales coverage remain unmatched.	Match the national balance Manish; Nigel source access	3 / 6
	S	Demand is concentrated, but later provincial data are estimates GP, KZN and WC account for 68.6% of reported 2022 sales. The map does not establish customer access.	Refresh observed demand and destinations Manish; Nigel review	4 / 5
Market changes	S/C	Demand drivers can move in different directions Power, freight, prices and vehicle mix need separate mechanisms; observed activity is not a fuel forecast.	Define growth, lever cases and back-tests Manish; Nigel review	7 / 9
	C	Supply change makes import needs conditional Capacity has declined; redevelopment is conditional. Secunda/Natref product output and future cases remain open.	Confirm plant status, timing and yields Manish; Henry review	8 / 9
	C	Alternative gateways challenge the same inland market Matola and Walvis Bay have storage; their competitiveness depends on full costs, route capacity and access.	Compare one product and destination Manish; Nigel / Henry	10 / 11
Vopak outlook	R	Market share of Vopak in Dbn and Lesedi Share is not established. Published tanks show footprint; unique customer deliveries establish served demand.	Obtain flows, remove transfers, test capture gates Nigel client data; Manish	12 / 13 / 14
	R	Potential scope for new investment? Prioritise Durban/Lesedi customer flows. Additional storage follows secured volumes and a usable-capacity gap.	Validate where, how and when to act Nigel; Manish / Henry	15 / 16 / 17

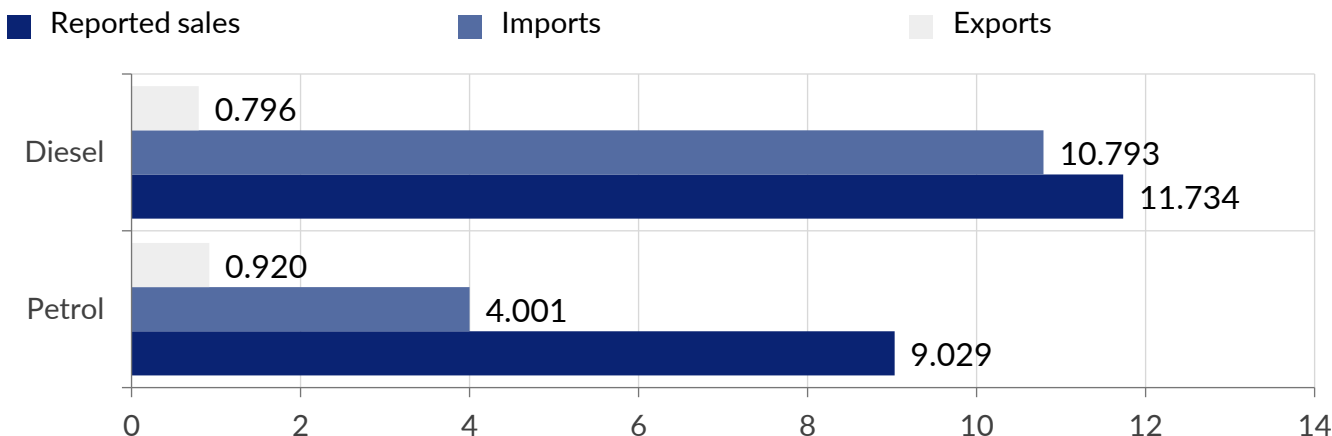
S = Situation C = Complication R = Resolution | Every Henry storyboard prompt has a response on pages 18–20; open items remain open.

Customs records 10.0 billion litres of net diesel imports in 2024

National petrol/diesel | 2024, billion litres

Production + imports – exports – stock build = consumption

Compare consumption with reported sales; explain coverage and the residual.



Net imports: petrol 3.081, diesel 9.997; combined 13.078 bn litres.

Primary trade | SARS customs

Diesel imports 10.793 bn L.
Government rounds to 10.800; FIASA 14.793.
Use SARS; retain the source discrepancy.

2024 accounting remains open

Sales less net imports = 7.685 bn L.
Balancing requirement, not production.
Production, stocks and coverage unresolved.

What the evidence shows

01 | Customs supports the trade selection

SARS records diesel imports of 10.793 and exports of 0.796 bn litres in 2024. Net imports are 9.997; use customs as the primary trade source.

02 | Reported sales remain a proxy

FIASA reports petrol/diesel sales of 20.763 bn litres for 2024. Editions differ; this demand proxy remains unverified and is not imports plus exports.

03 | Production and stocks remain open

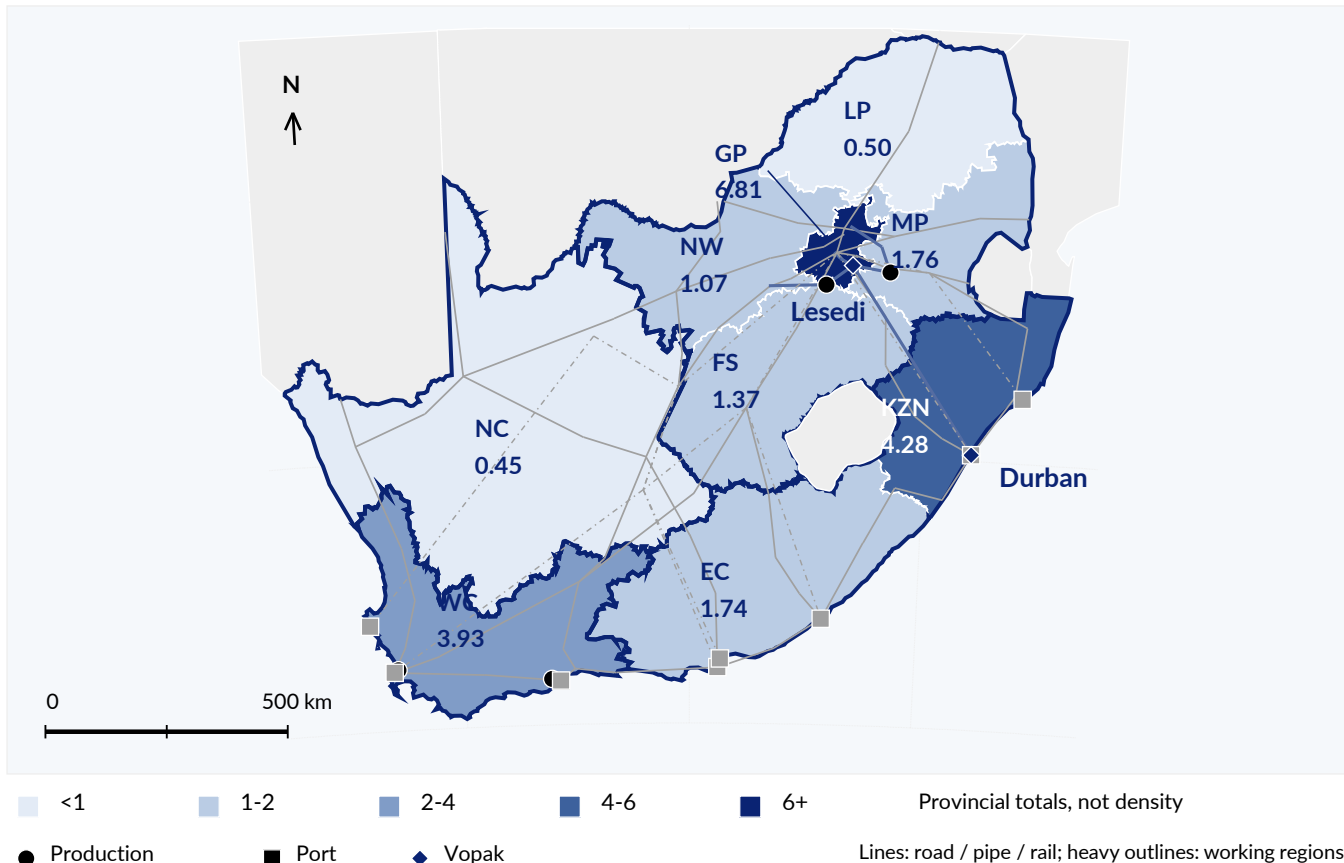
Production and stocks remain unresolved. The proposed 3-4 bn litre diesel gap is a hypothesis, not measured consumption.

To resolve next

Manish: rebuild the balance CSV from SARS; match product, year, actual output and stock changes. Nigel: agree the sales selection and obtain customer/port evidence.

Three provinces account for 69% of reported fuel sales

Petrol + diesel sales by province | 2022, bn litres/year



What the evidence shows

01 | Demand is concentrated

Gauteng, KwaZulu-Natal and Western Cape account for 68.6% of reported 2022 petrol/diesel sales.

02 | Four reporting regions

Eastern coast: EC/KZN. Inland: GP/FS/LP/MP/NW. Western coast: WC. Other: NC. Boundaries are reporting groups.

03 | Sales locate the historical market

Colours show provincial sales totals, not density or customer catchments. Only Durban and Lesedi are labelled; competitor detail is on pages 13 and 15.

To resolve next

Nigel: confirm customer markets for Durban and Lesedi. Manish: retain the observed 2022 map; present later estimates separately with national-source flags.

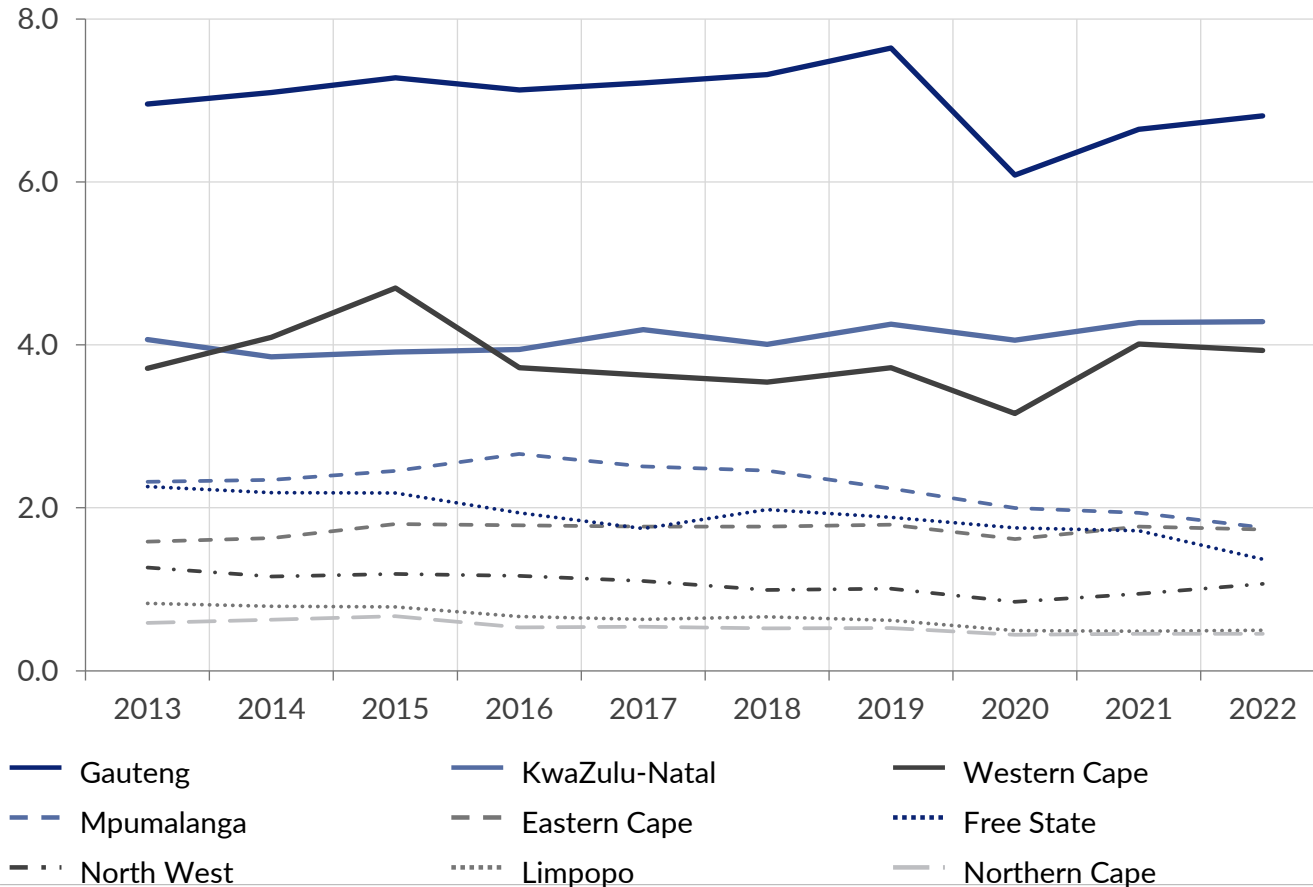
Sales totals, not density. Vopak locations highlighted; detailed operator inventory on pages 13 and 15. Jet excluded.

Source: DMPR 2022 sales; geoBoundaries/OCHA/MDB 2020 provinces (CC BY 3.0 IGO); operator sources on slide 12.

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Provincial sales show different demand patterns over time

Annual provincial sales | 2013–2022, bn litres/year



What the evidence shows

01 | Gauteng remains the largest market

The reported series places Gauteng above the other provinces throughout 2013–2022. Volumes differ across the smaller markets.

02 | Later provinces are estimates

2024 petrol estimates: Gauteng 3.47, KZN 1.47, Western Cape 1.41 bn L. These hold 2022 shares; no later observed provincial sales were found.

03 | Historical differences are traced

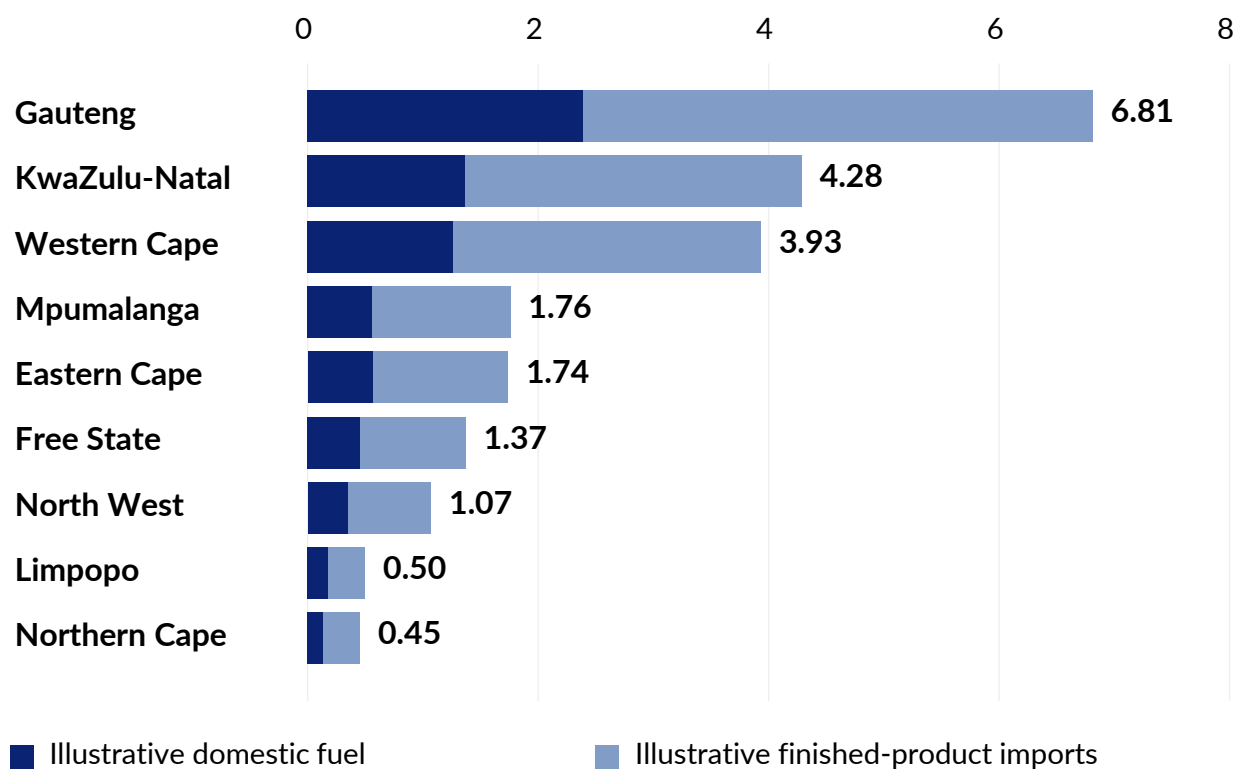
The 2013 national parser is corrected on the analyst branch. 2014/2015 source versions differ; 2018 district coverage is incomplete. Preserve the flags.

To resolve next

Manish: reconcile source selections and retain estimate labels. Nigel: decide whether estimated petrol volumes belong in the outlook; diesel booking shares are less stable.

Provincial demand is known; its supply origin still needs evidence

2022 sales totals | ILLUSTRATIVE domestic/import allocation



Units: bn litres/year. Imported/domestic segments are a scenario illustration, not measured provincial flows.

What the evidence shows

01 | Demand totals are sourced

The provincial bars use reported 2022 petrol/diesel sales.

02 | Origin shares are illustrative

Domestic/import fractions are authored examples; no actual provincial origin allocation is established.

03 | Imports need a national balance

Exports and stock movements must be reconciled before a sourced import requirement can be presented.

To resolve next

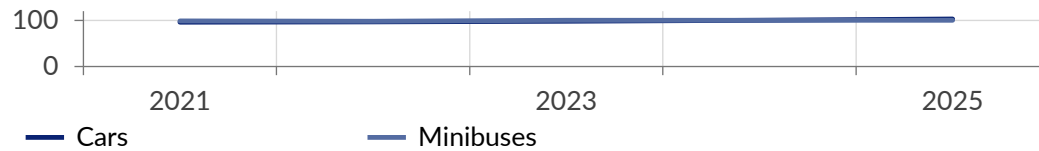
Manish: populate the matched-year national balance first; retain provincial origin as unknown until supported.

Observed activity helps define the fuel levers; intensities remain open

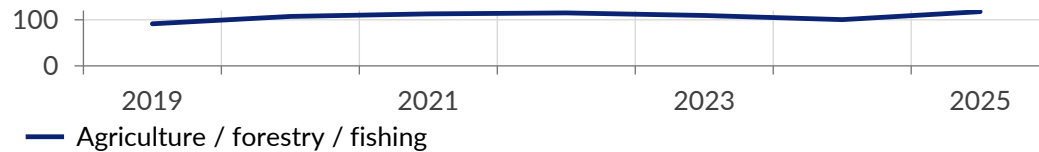
Potential demand levers | 2024 = 100; price panel uses nominal R/litre

Observed proxies, not fuel-volume effects. Activity axes 0–120; EV 0–400; prices 0–30 R/L.

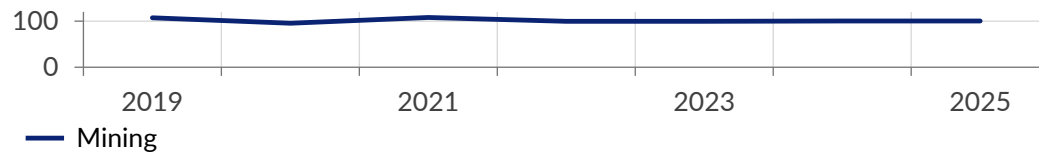
Passenger travel | all-fuel stock, Dec



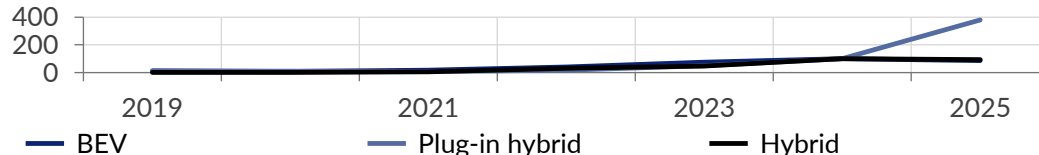
Agriculture | real value added, CY



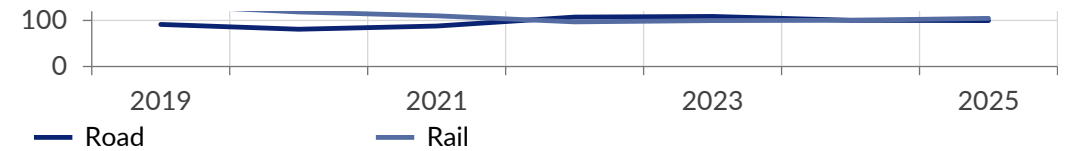
Mining | production volume, CY



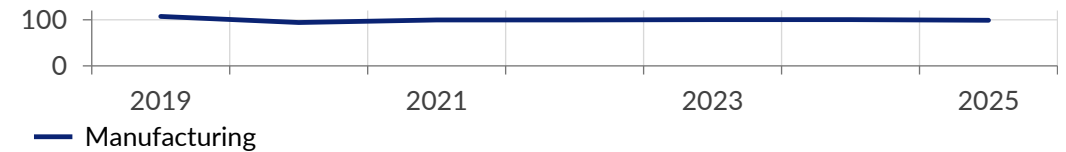
Electrification | annual new sales, CY



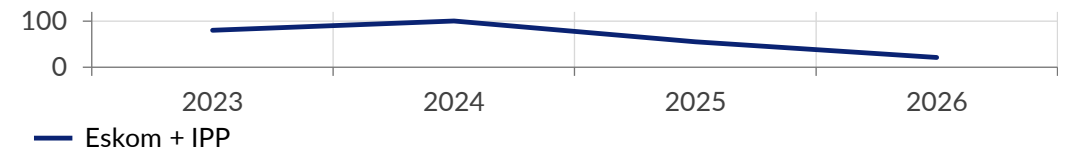
Freight demand | payload, CY



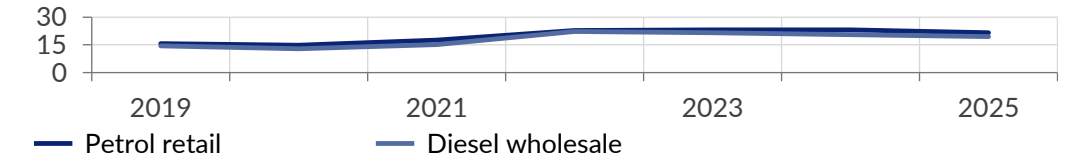
Manufacturing | production volume, CY



Diesel power | OCGT generation, FY



Price response | annual nominal R/litre



Latest staged: activity Jul 2026; GDP Q2 2026; inland prices Oct 2026. Charts use complete years. Fuel intensities remain open; FY ends March.

Source: NaTIS; Stats SA P0441 / P2041 / P3041.2 / P7162; Eskom; naamsa; DMPR / CEF. Snapshot manifest in notes.

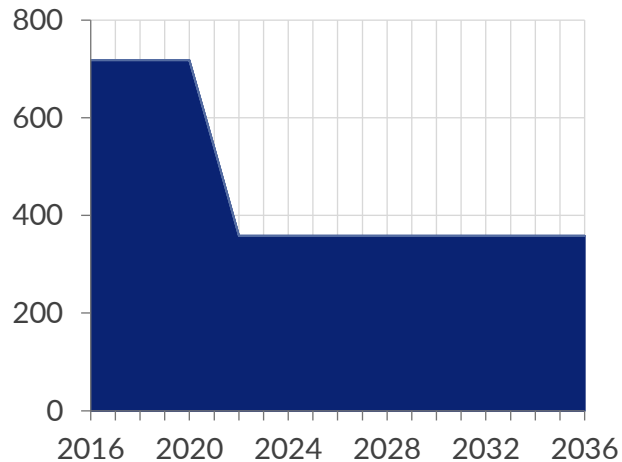
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Refinery capacity has fallen; redevelopment remains conditional

Published footprint + illustrative scenarios | thousand bbl/day

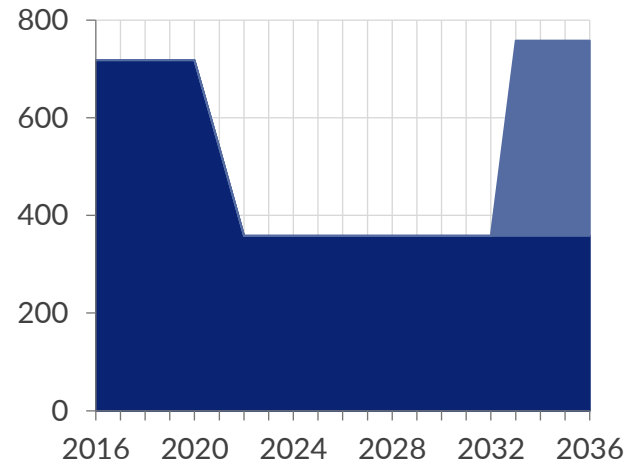
History 2016–2025 is sourced; 2026 onward is illustrative, not a forecast

A | No new refinery capacity



2036: 358 thousand bbl/day

B | Conditional CEF redevelopment



2036: 758 thousand bbl/day

Reported FY2024 output: Secunda 29.1; Natref Sasol share 17.8 million barrels.

All products; year to June, not calendar-year petrol/diesel. B adds 400 from 2033, conditional on FID.

What the evidence shows

01 | Capacity has fallen

Published capacity falls from 718 to 358 thousand bbl/day between 2020 and 2022. Redevelopment remains conditional.

02 | Reported output is different

FY2024 Secunda output is 29.1 million barrels; Natref is 17.8 for Sasol's share. Both cover all products and a year to June.

03 | A product bridge is needed

Match product yields, calendar years and downtime before calculating import effects. Capacity does not measure actual output.

To resolve next

Manish: match operator output to product/year and document Natref ownership basis. Henry: review technical mechanisms. Nigel: agree refinery scenario timing.

Connect historical changes to forecast cases and market decisions

Market changes | historical evidence, forecast cases and decision outputs

STRUCTURE FOR REVIEW | assumptions and fuel-volume effects remain uncalibrated

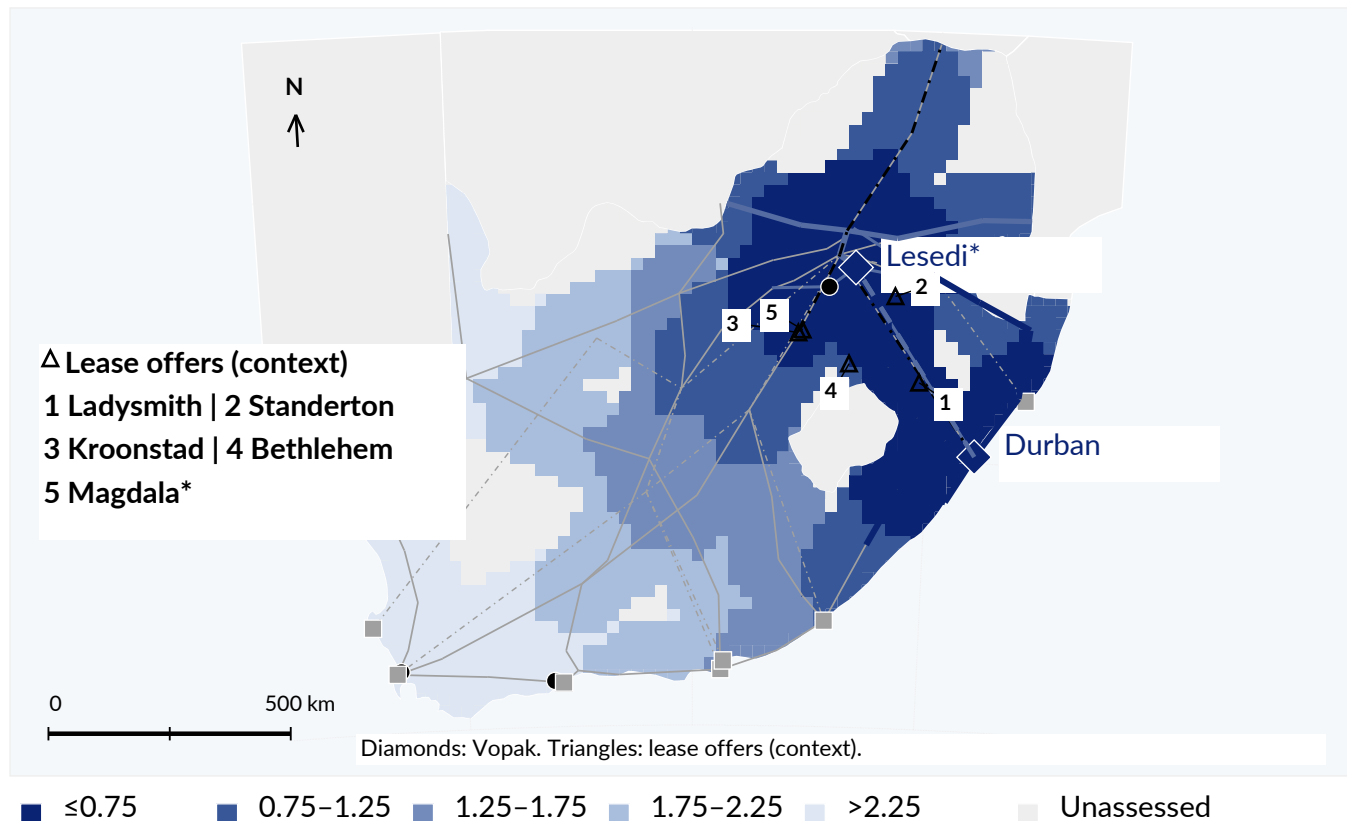
Historical evidence	Forecast cases to define	Output to calculate	Decision it supports
Fleet, activity and prices Observed history and recent partial periods	Base: observed performance continues. High / low: explicit growth, mileage and price assumptions	Petrol / diesel demand by year; forecast bands and sensitivity	Which customer markets grow, contract or change product mix?
EV, freight, sectors and power Separate activity from fuel use	Change uptake, modal share, fuel intensity and OCGT use; separate BEV and hybrids	Incremental litres by lever; avoid overlap with the base growth path	Which service proposition and customer segment should Vopak prioritise?
Refinery capacity and reported output Separate actual from conditional additions	Existing plants: output, downtime and yields. Redevelopment: conditional timing and ramp-up	Product-specific domestic output; matched imports, exports and stock changes	When could additional import handling or storage be needed?
Provincial sales and competing routes Later fuel shares are estimates	Allocate national cases with explicit regional assumptions; test route cost and access	Reachable customer litres by region and route; actual Vopak deliveries separately	Where to play, how to win and the conditions for committing capacity

Historical evidence → agreed assumptions and levers → annual demand and supply → regional access → Vopak decision

History is observed; forecast cases are proposed. No new fuel forecast is shown until the assumptions, source selections and back-tests are reviewed.

Transport cost shapes the market Durban and Lesedi can reach

Illustrative road cost, R/litre



Example rate: R0.15/L dispatch + R0.002/L/km. Grey = unassessed.

Illustrative road cost only. Pipeline/rail access and commercial rights are not priced.

Source: Existing appendix routes; illustrative candidate connections, not verified catchments. Natural Earth 1:50m.

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What the evidence shows

01 | Darker means lower transport cost

Cells show example road cost from Durban or Lesedi. The rate is illustrative and excludes fuel price, storage and pipeline tariffs.

02 | Grey means unassessed

Grey areas have no assessed road connection under this illustration. They are not confirmed to be inaccessible.

03 | A cost surface is a first screen

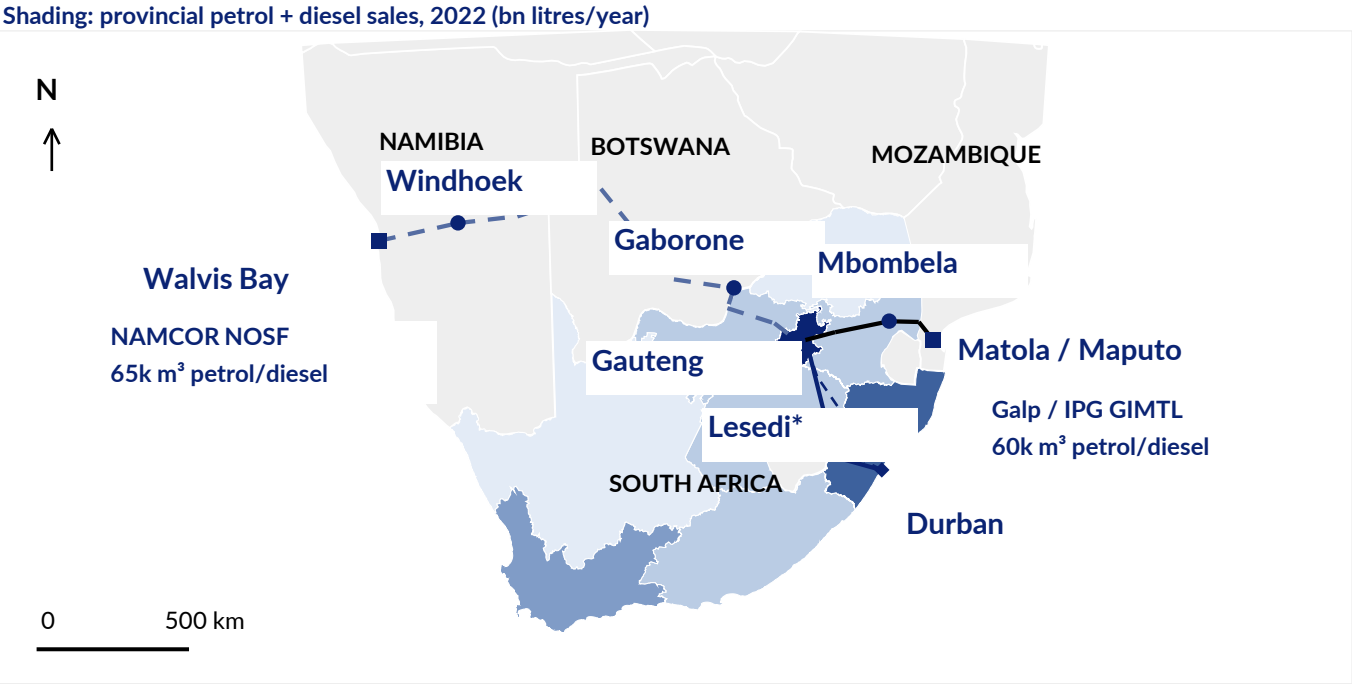
Pipeline and rail lines are context. Route capacity, product compatibility and customer rights still determine serviceable demand.

To resolve next

Nigel: select the first customer destination and cost threshold. Manish: replace example rates with dated route costs and access evidence.

Compare Durban, Matola and Walvis Bay for the same inland customer

Durban, Matola and Walvis Bay | schematic corridor comparison



What the evidence shows

01 | Vopak has a South African footprint

Durban and Lesedi are highlighted against 2022 provincial sales. Colour shows sales totals; it does not establish Vopak market dominance.

02 | Matola has published fuel storage

Galp/IPG GIMTL lists 60,000 m³ of petrol/diesel tanks, plus 6,000 m³ LPG excluded here. N4 is the eastern inland comparison route.

03 | Walvis has storage and handling

NAMCOR NOSF lists 75,000 m³ in total, including 65,000 m³ petrol/diesel. Trans-Kalahari is the western inland comparator.

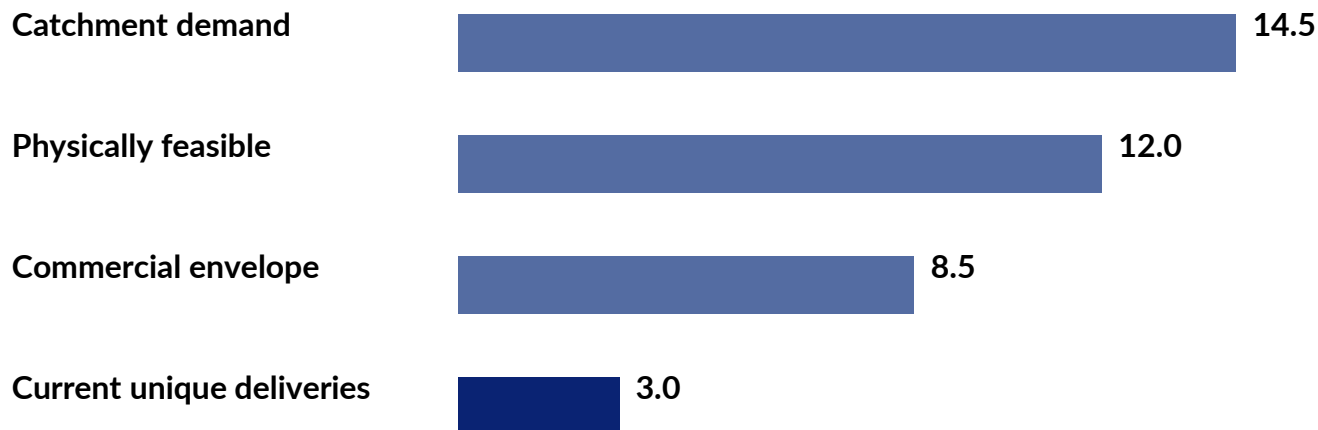
To resolve next

Manish: verify available tanks, customer access and full delivered costs. Source Namibia/Mozambique demand before extending the heatmap; grey means unassessed.

Separate market demand, accessible volume and unique Vopak deliveries

Market screens | illustrative bn litres/year

Successive screens; these volumes are illustrative



Count Durban–Lesedi transfers once

Durban 2.8 + Lesedi 2.0 – shared transfer 1.8 = unique 3.0

Demand ≠ terminal receipts ≠ storage capacity. Incremental opportunity is on page 14.

What the evidence shows

01 | Count customer demand once

Example receipts of 2.8 at Durban and 2.0 at Lesedi include a shared 1.8 transfer. Unique demand served is therefore 3.0 bn litres.

02 | Accessibility precedes opportunity sizing

Example catchment demand 14.5 → physically feasible 12.0 → commercial envelope 8.5. Each screen requires evidence.

03 | Examples are separate from sales

These market volumes illustrate the calculation. They are not the reported 2022 regional totals or actual Vopak deliveries.

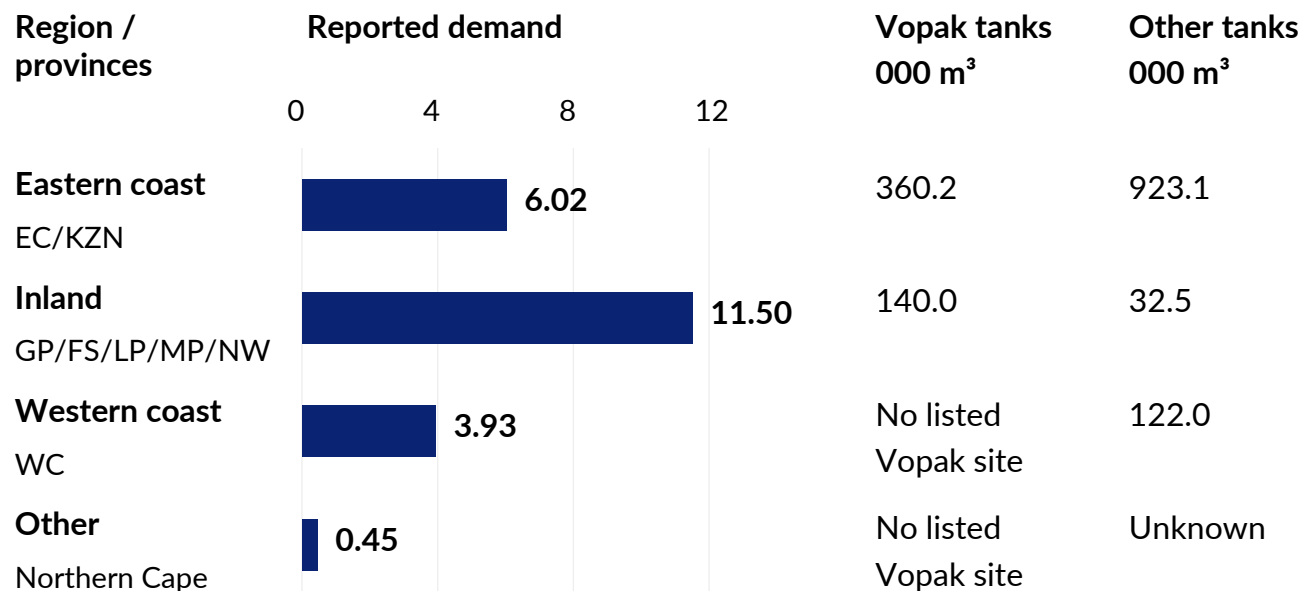
To resolve next

Nigel: request customer deliveries, destinations and transfer records. Manish: calculate unique served demand before estimating additional access.

Compare regional demand with Vopak's published storage footprint

All four regions | reported 2022 demand, bn litres/year

National total 21.90 bn L | petrol + diesel; jet excluded



Published capacity is a provisional footprint measure, not annual demand served.

Partial gross inventory; mixed products and unknown Sasol/Transnet capacities.

Fuel-compatible capacity share and actual Vopak demand share remain unestablished.

[1] Vopak [2] Bidvest [3] Burgan [4] Shell [5] NERSA [6] Sasol [7] Transnet

Source: DMPR 2022 provincial petrol/diesel sales; public operator sources [1-7], checked 6 Oct 2026. Share not established.

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Regional competitor evidence

01 Eastern coast

Vopak: Durban [1]. Bidvest: Durban and Richards Bay [2]; mixed-product capacity.

Transnet: Ladysmith lease tanks [7].

02 Inland

Vopak: Lesedi [1]. Bidvest: Isando [2].

Sasol: Alrode, Pretoria West, Waltloo, Sasolburg [6].
Transnet: Tarlton / Jameson Park + four lease sites [7].

03 Western coast

Burgan Cape: Cape Town petrol/diesel terminal [3].
Confirm current tank availability and customer access.

04 Other / Northern Cape

Operator coverage incomplete. Check Shell site notices [4] and NERSA licences/access records [5].

The illustrative additional opportunity is 5.5 billion litres a year

Accessibility gates and volume milestones | illustrative bn L/year

ILLUSTRATIVE | each increment requires evidence; not cumulative terminal receipts

5.5 bn L/year additional candidate

3.0 current example + 5.5 additional = 8.5 total envelope



What must hold before any additional litres are counted?

Competitive cost Usable capacity Customer access Unique deliveries

Detailed customer-by-route gates belong in the evidence register.
Illustration only; 5.5 is neither a forecast nor a confirmed capturable market.

What the evidence shows



01 | Reach is necessary, not sufficient

Accessibility map p10 and competing routes p11 locate candidate access. Neither currently links a verified cost threshold to customer litres.

02 | Example milestones are conditional

The current example is 3.0. Coastal candidate 1.5 raises it to 4.5; inland candidate 4.0 raises it to 8.5. These are authored volumes.

03 | Every increment needs four gates

Evidence cost, route/tank capacity, customer rights and unique deliveries for the same product and period. Unassessed litres remain explicit.

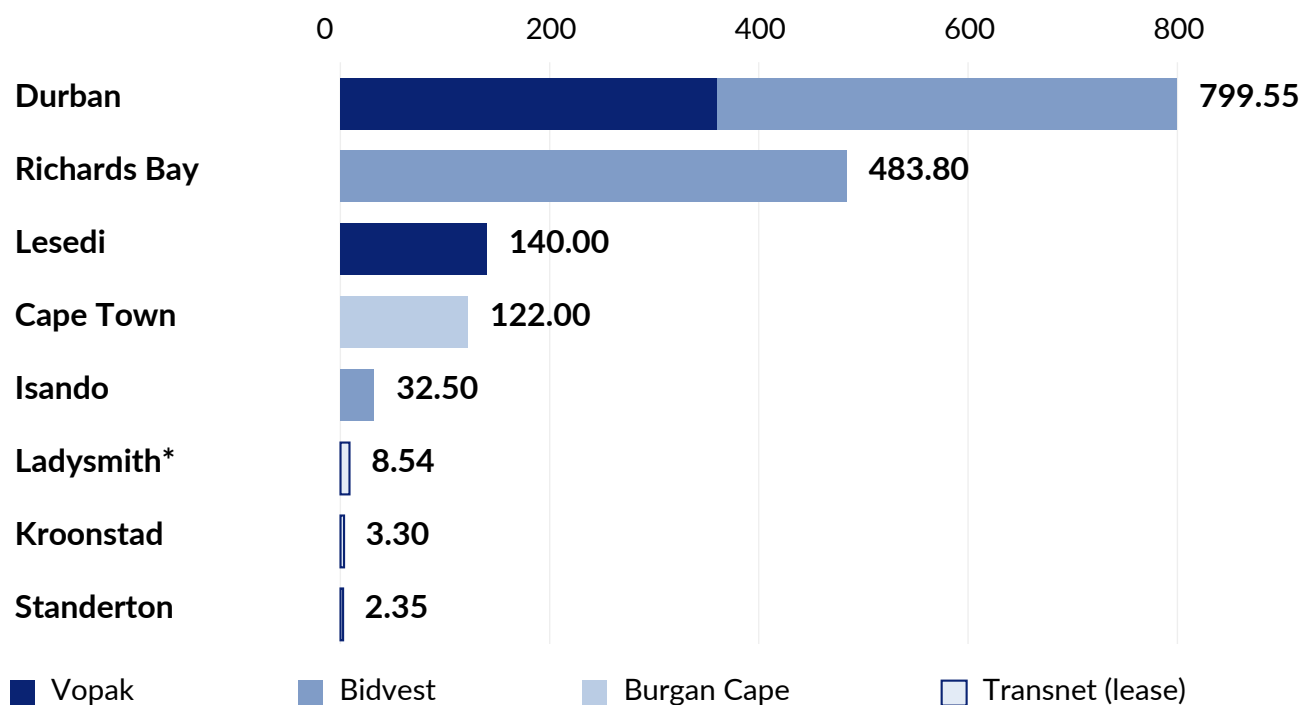
To resolve next

Nigel: agree destinations and acceptable delivered cost.
Manish: build a customer-by-route gate table; record passed, failed and unassessed litres, with sources and conditions.

Durban leads the published storage subset

Published storage by location | thousand m³; mixed capacity bases

One shared scale for every location



Small lease bars reflect smaller tank volumes; they are not operating supply.

*Ladysmith excludes 0.474 intermixture. Unquantified: Bethlehem, Magdala, Sasol, Tarlton, Jameson Park.

Source: Vopak; Bidvest; VTTI Burgan Cape; Transnet lease RFPs (5 Oct 2026), pp. 2-3. Checked 6 Oct 2026.

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What the evidence shows

01 | Capacity is a stock

Published tank volumes are not annual fuel deliveries or market share.

02 | The inventory is partial

Missing capacities are unknown, rather than zero; mixed product scopes limit comparison.

03 | Leases are conditional

Transnet lease tanks are not counted as currently available operating supply.

To resolve next

Manish: verify product compatibility, capacity basis and operating status. Nigel: confirm usable client capacity.

Prioritise customer markets, win on service and cost, then expand

Market outlook | priority hypotheses and conditional timing

Where to play	How to win	When to act	What must be true
Durban coastal customers First market to test	Competitive delivered cost; reliable receipt and dispatch; compatible tanks	Now: identify customers and test service using existing assets	Named customer, product and unique litres; cost threshold; usable tank and dispatch space
Lesedi inland customers First inland market to test	Reliable inland supply; competitive full-route cost versus other gateways	After route and customer validation: negotiate incremental flows	NMPP or alternative route access; receipt/dispatch capacity; customer rights and switching
Matola and Walvis Bay Competing gateways to compare	Test storage, handling and transport to the same inland destination	During the inland comparison; pursue an opportunity only if access and economics are evidenced	Available compatible tanks; border/route feasibility; partner and customer access; full delivered cost

Investment gate

New storage follows secured incremental flows and a gap in usable capacity; the illustrative 5.5 bn litres does not establish a commitment.

Nigel: Choose the first customer/product and acceptable cost. Confirm the service proposition and commercial access.

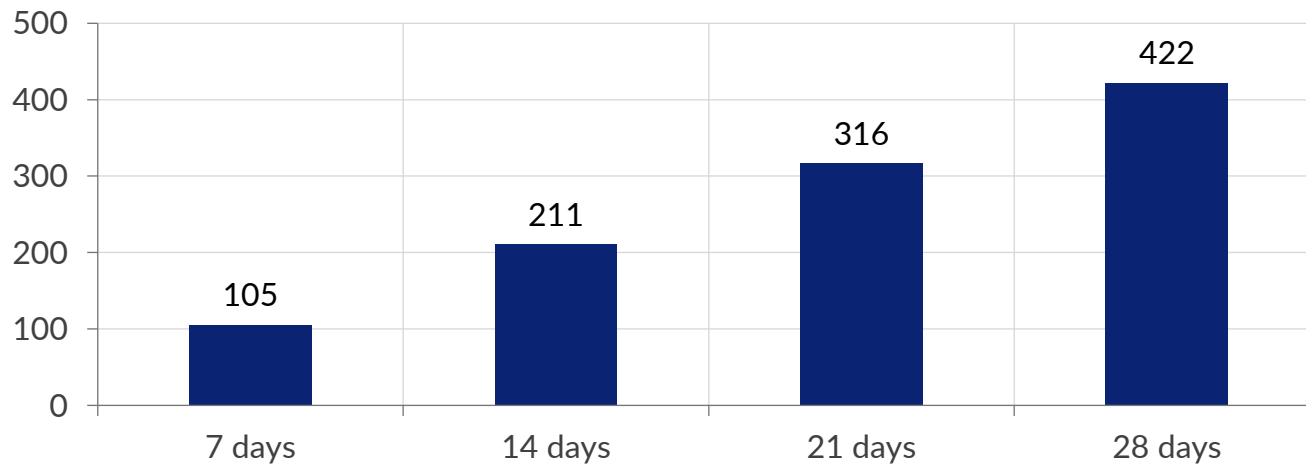
Manish: Build a customer-by-route evidence table: litres, cost, capacity, rights, timing and owner. Mark each gate passed, failed or unassessed.

Translate additional fuel flows into a preliminary inventory requirement

Working inventory sensitivity | illustrative, thousand m³

ILLUSTRATIVE | additional 5.5 bn litres/year; uniform daily flow

More inventory days require more working stock



Working m³ = annual litres ÷ 365 ÷ 1,000 × inventory days

At 14 days: approximately 211 thousand m³ of working inventory.

New tanks depend on usable spare capacity, peaks, product segregation and dispatch limits.

What the evidence shows

01 | Inventory days set working stock

At the illustrative 5.5 bn litres/year, 14 days needs about 211 thousand m³. More inventory days require more stock.

02 | Existing tanks may cover part

Usable spare tanks may absorb this inventory. Published gross capacity does not establish spare petrol/diesel space.

03 | Sizing needs an operating case

Test peak receipts, product segregation, tank heels and dispatch limits. This is a sensitivity, not an investment recommendation.

To resolve next

Nigel: confirm candidate flows and inventory policy.
Manish and Henry: test usable spare tanks, peak flows and operating allowances before sizing new capacity.

Henry storyboard answers | Market baseline

S | Every prompt addressed; partial and open findings require the stated evidence

Henry storyboard prompt	Current answer / evidence gap	Deep dives
Total SA demand development	Partial: reported 2024 sales are 20.763 bn L; coverage awaits reconciliation. Provincial history ends in 2022.	3 / 4 / 5
Eastern SA market within economic striking distance	Open: location and illustrative costs show reach, not accessible customer litres. Need destinations, tariffs and rights.	4 / 10 / 12 / 14
Greater diesel use in power generation	Partial: generation history is shown; recent OCGT use falls. Diesel-litres attribution still needs fuel-use evidence.	7 / 9
Rail to road	Partial: road/rail payload history is shown. Payload alone does not prove modal substitution or diesel effects.	7 / 9
Changes in fuel price	Partial: complete-year price history is staged. Travel response and real-price effects remain uncalibrated.	7 / 9
More fuel-efficient vehicles	Open: fleet stocks and new EV/hybrid sales do not establish fleet efficiency. Need fuel mix, mileage and intensity.	7 / 9
Slow GDP growth	Partial: GDP and sector activity evidence is available; growth-to-fuel effects require explicit assumptions.	7 / 9
Refinery closures	Partial: published capacity falls to 358 thousand bbl/day. Capacity is distinct from output and operating availability.	8
Challenges at Secunda	Partial: FY2024 output is 29.1 million barrels, all products. Availability and petrol/diesel yields remain open.	8 / 9
Closure of Moss gas	Open: plant-specific shutdown dates, operating status and lost product output need explicit source confirmation.	8 / 9
Growth in imports and where they enter	Partial: 2024 net imports are 13.078 bn L. Import-growth attribution, product-specific ports and origin remain open.	3 / 6

Henry storyboard answers | Market changes

C | Every prompt addressed; partial and open findings require the stated evidence

Henry storyboard prompt	Current answer / evidence gap	Deep dives
Road to rail; private-sector rail projection	Open: historical payload is available; private-sector service, timing and modal-shift assumptions need evidence.	7 / 9
End of loadshedding and spare power capacity	Partial: observed OCGT generation falls. Future dispatch, spare capacity and diesel demand remain conditional.	7 / 9
EV / HEV penetration projection	Open: sales uptake is shown, not fleet penetration. Separate BEV, PHEV and conventional hybrids; model turnover.	7 / 9
Effect of EV / HEV on litres per km	Open: require stock mix, mileage and fuel intensity. BEVs and hybrids have different fuel-use mechanisms.	7 / 9
Higher fuel prices and km travelled	Open: prices are sourced; mileage response and elasticity must be specified and tested before forecasting.	7 / 9
Mega refinery / SAPREF restart	Conditional: redevelopment capacity is a scenario, not sanctioned output. Confirm FID, ramp-up and product yields.	8 / 9
Secunda gas, coal and MRG diversion to 2032	Open: no liquid-fuel displacement is inferred. Need dated gas allocation, plant mechanism and product-output evidence.	8 / 9
Natref future?	Open: reported FY output is available; future ownership, availability and product yields require a dated case.	8 / 9
Durban storage / NMPP / Lesedi vs road and rail	Open: compare the same customer and product. Need full delivered costs, capacity, tariffs and service availability.	10 / 11 / 14
Transport from Matola and Walvis Bay	Partial: gateways, published storage and schematic routes are shown. Full costs, borders and customer access remain open.	11 / 14 / 16

Henry storyboard answers | Vopak outlook

R | Every prompt addressed; partial and open findings require the stated evidence

Henry storyboard prompt	Current answer / evidence gap	Deep dives
Market share of Vopak in Dbn and Lesedi	Open: published storage is an asset footprint, not served demand. Need unique deliveries, transfers and matched demand.	12 / 13 / 14
Potential scope for new investment?	Conditional: prioritise customer flows and service first. New storage requires secured incremental flows and a usable-capacity gap.	15 / 16 / 17

Agree priorities and next steps

Confirm demand, routes and access

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Lesedi | vopak.com



Presentation palette | colours and HEX codes

Reference | colours used in this presentation

Core presentation colours

	Vopak blue #0A2373		Secondary blue #546CA2		Text / ink #000000		White #FFFFFF
	Unassessed / neutral #EEEEEE		Header divider #A0A0A0		Body divider #B8B8B8		Frame divider #B0B0B0

Sales-map colours | provincial totals, bn litres/year

	Below 1 #E3EBF6		1 to below 2 #BACCE4		2 to below 4 #809CC7		4 to below 6 #3D619D
	6 and above #0A2373		Map background #F5F8FB				

Other template theme colours

	Theme dark 2 #11151A		Theme light 2 #E7E6E6		Accent 3 #BDBEC1		Accent 4 #404040
	Accent 5 #767676		Accent 6 #D9D9D9				

Typeface: Lato. Sales colours show totals, not density or market share. Grey denotes unassessed demand.

Source: supplied Vopak template; pptx/brand_configs/vopak.py; provincial_demand_map.py.

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